

AegisOrbit - Technical Documentation

Team: RM99781

Public URL: <https://app-aegis-orbit-rm99781.azurewebsites.net>

GitHub repository: <https://github.com/pebarone/aegis-orbit>

Primary SDG: ODS 9 - Industry, Innovation and Infrastructure

Problem: LEO satellite operators and SSA teams need fast, explainable support during orbital conjunction

Solution: AegisOrbit is a FastAPI Space Situational Awareness MVP. It simulates deterministic orbital catalog

Architecture: FastAPI + Uvicorn on Azure Linux App Service. Deterministic simulator feeds landing, dashboard,

Azure resources: rg-aegis-orbit-rm99781, asp-aegis-orbit-rm99781, app-aegis-orbit-rm99781, law-aegis-orbit-

Security: No secret values in source. GitHub Actions uses AZURE_CREDENTIALS secret. Key Vault secret exists

Monitoring: Application Insights provisioningState Succeeded. Alert rule enabled, severity 2, metric Http5xx,

CI/CD: Workflow [.github/workflows/deploy.yml](#) runs on push to main. GitHub repository secret AZURE_CREDENTIALS

Validation: Local pytest passed: 6 passed. Public smoke tests passed for /health, /dashboard, /docs, and

[Landing](#) [Dashboard](#) [API Docs](#)

AegisOrbit

Purpose

AegisOrbit is a Space Situational Awareness MVP for satellite operators and SSA teams. It simulates orbital catalog ingestion, conjunction risk scoring, and evasion-routing recommendations for LEO assets.

Team

Team identifier: **RM99781**.

Problem

Growing satellite constellations and debris increase collision risk, fuel waste, and Kessler Syndrome exposure. Operators need fast, explainable maneuver support.

ODS 9

Protects critical orbital infrastructure through resilient software, monitoring, and realistic cloud deployment on Azure App Service.

AegisOrbit Operations Dashboard

Deterministic simulated conjunction alerts for LEO assets. Data is fixed for demos and tests; no external API required.

AEO-LEO-104

Conjunction object: `DEBRIS-1998-067QZ`
Event: 2026-06-03T14:35:00Z

0.42 km miss distance

12.8 km/s relative velocity | Pc 0.00031
Severity: **critical**

AEO-LEO-118

Conjunction object: `ROCKET-BODY-2015-021B`
Event: 2026-06-04T02:10:00Z

1.9 km miss distance

9.4 km/s relative velocity | Pc 8e-05
Severity: **watch**

AEO-IOT-022

Conjunction object: `DEBRIS-2020-061AF`
Event: 2026-06-04T19:45:00Z

4.6 km miss distance

7.1 km/s relative velocity | Pc 1.5e-05
Severity: **nominal**

AegisOrbit SSA API

0.1.0

OAS 3.1

/openapi.json

Deterministic Space Situational Awareness MVP for orbital conjunction monitoring and evasion routing.

default



GET	/	Landing	▼
GET	/dashboard	Dashboard	▼
GET	/health	Health	▼
GET	/api/status	Status	▼
GET	/api/conjunctions	Conjunctions	▼
POST	/api/evasion-routing	Evasion Routing	▼

Schemas



EvasionRoutingRequest	Expand all	object
HTTPValidationError	Expand all	object
ManeuverRecommendation	Expand all	object
ValidationError	Expand all	object